Project Initialization and Planning Phase

Date	14 July 2026
Team ID	
Project Name	Automated Credit Card Application Screening using ML
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) Report

This project automates the credit card screening process using advanced machine learning. By analyzing financial profiles, employment duration, and historical credit behavior, the system predicts approval outcomes in real-time. It empowers bank analysts and compliance officers with model explainability (SHAP) while offering customers a sandbox to test application scenarios, ensuring a transparent and highly efficient financial workflow.

Project Overview	
Objective	Build a cloud-deployed ML application to automate credit card screening with high precision, utilizing hyperparameter-tuned classification models to facilitate real-time decision-making.
Scope	Includes data preprocessing (handling multi-class payment status), feature engineering (scaling/ normalization), training four comparative models (Logistic Regression, Random Forest, XGBoost, Decision Trees), and deploying via IBM Watson Machine Learning with a Flask dashboard.
Problem Statement	
Description	Traditional credit screening relies on manual verification which is slow, prone to human error, and lacks the scalability to handle thousands of modern-day applications daily.
Impact	Delays in approval lead to customer churn, while inconsistent risk assessment increases the bank's exposure to "high risk" defaults and operational overhead.
Proposed Solution	
Approach	Comparison of four tuned algorithms using Scikit-learn. The best model is deployed on IBM Cloud. A Flask interface provides real-time predictions, SHAP-based explainability, and a notification system for manual review of borderline cases.
Key Features	• Real-time automated approval/rejection predictions • Multi-model comparison (XGBoost, RF, LR, DT) • Model Explainability (SHAP values) for compliance • Interactive Dashboard (Income vs. Approval visualization) • Borderline Case Notification & Customer Sandbox Mode • Scaleable deployment on IBM Watson Machine Learning

Resource Requirements

Resource Type	Description	Specification/ Allocation
Hardware		
Computing Resources	CPU/Cloud Compute	Intel Core i5/i7 (4+ Cores) / IBM Cloud VCAPs
Memory	RAM requirements	8 GB RAM (Minimum), 16 GB (Recommended)
Storage	Cloud/Local Storage	512 GB SSD (Local) / Cloud Object Storage
Software		
Frameworks	Python Web Framework	Flask (v3.0+), Jinja2
Libraries	Data Science Suite	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, SHAP
Deployment	Cloud Platform	IBM Watson Machine Learning, IBM Cloud Pak
Dev Environment	IDE	VS Code, Jupyter Notebooks, Git/GitHub
Data		
Data Source	Financial Datasets	Kaggle/UCI Credit Card Approval Datasets (~30k records, CSV)